Study Buddy

Product Requirements Document: Net New Build

Build name: Study Buddy

Owners: Ivaylo Nanev and Soma Majumder

Date: October 21, 2026

Delivery status — October 26, 2026

Status is measured against merged dev commit 87c800e. Implemented means backend code merged into dev and checked offline; it does not mean deployed, independently proven content quality, or completion of the student pilot. The product requirements below describe the intended experience; P0/P1/P2 indicate priority, not delivery status.

Implemented in dev: approved lesson loading (currently one cell-membrane lesson, version 2); Tutor teaching and focused reteaching; Assessment quiz generation and verified-result explanations; deterministic controller grading; a separate Reviewer task, one bounded content revision and exact-candidate publication; one learning retry; durable attempt/feedback/weak-concept records; final-summary persistence with pending-save recovery at the finish_session level; learner-scoped ownership checks; identical-submission replay and changed-answer rejection.

Implemented operations: explicit opt-in for paid Claude calls; crash-expiry cleanup; a foreground maintenance worker with interruptible shutdown and three temporary-lock retries (1, 2 and 4 seconds); migration discovery, verified baseline adoption through 008, backups and atomic pending-migration execution. These maintenance tools are not a deployed service. PRs #39, #40 and #43 are merged.

Under review: PR #45 adds resume_summary for retrying the final save using the existing session ID, without a new session or agent calls. Until merged, this convenience action is available only on codex/summary-recovery; the existing finish_session primitive already supports retrying pending saves.

Local prototype, not merged: codex/lesson-picker contains a terminal demo with lesson selection, a saved hand-written Tutor explanation and quiz, answer collection and deterministic grading with saved feedback. It makes no live AI calls or persistent progress writes. This is not the production web interface or a live-agent demonstration.

Planned or not yet verified: the student web interface and authentication method; expansion from one lesson to the five-lesson pilot; returning-learner history/recommendation experience; P1 confidence ratings, misconception evidence, spaced review and source-linked progress; production hosting and maintenance supervision; independent live Reviewer evaluation and the student pilot. Offline tests establish workflow mechanics, not learning outcomes or factual quality.

Ownership: Ivaylo leads controller orchestration, persistence, recovery and operational tooling. Soma leads Assessment quiz/feedback behavior, quality rubric and benchmarks. Shared contracts and workflow changes are reviewed together. The Reviewer is an additional isolated workflow role alongside Tutor and Assessment; the controller owns approval and all writes.

Document control: this Google PRD is the product source of truth. docs/Study-Buddy-PRD.pdf is its dated export; README.md summarizes the project and backend/README.md documents current operation. These are manually synchronized snapshots, not an automatic sync. When a PR merges or scope changes, update the status here, refresh the PDF and update both READMEs together.

1. PROBLEM

Beginner students studying independently can struggle to understand educational text and identify what they have misunderstood. Reading alone does not provide targeted feedback, so students may repeat material they already understand while missing important gaps. This is a problem hypothesis to validate with student testing.

1a. Opportunity

Turn approved educational lessons into a guided cycle of explanation, practice, feedback, and targeted reteaching. The project also gives its builders practice with tool use, structured agent handoffs, conditional routing, and persistent learning history.

Market Opportunity

- Initial opportunity: a pilot for beginner students using one subject and five curated lessons. Market size and student time savings have not been measured; validate demand through student interviews and a 10-session pilot.

1b. Users & Needs

Primary users: Beginner students who want to understand a supported topic and check their understanding through a short quiz.

Secondary users: People who curate lessons, review generated content, and inspect workflow failures.

Key User Needs

As a beginner student, I need a short explanation grounded in a visible source because the original material can be difficult to understand.

As a returning student, I need feedback and practice focused on my weak concepts because a score alone does not tell me what to study next.

2. PROPOSED SOLUTION

Study Buddy is a planned web app whose backend uses Tutor (Agent 1) for teaching, Assessment (Agent 2) for quizzes and result explanations, and a separate Reviewer task for quiz quality checks. After topic selection, the Tutor explains an approved lesson. Assessment creates ten questions and a private key; the controller validates the quiz, dispatches review and any single allowed content revision, and publishes only the exact approved candidate. The controller grades student answers deterministically and derives weak concepts; Assessment explains those verified results. Below 8/10, Tutor reteaches the missed concepts before one focused 10-question retry. The controller saves progress and the final summary before marking completion. See Delivery status for what is merged, under review and planned.

2a. Value Proposition

Beginner students use Study Buddy to check understanding, identify likely misconceptions, and revisit concepts across sessions without managing AI prompts. The core session ties every quiz to what was taught. The planned expansion combines answer confidence, misconception-aware feedback, spaced practice, and source-linked progress. Product promise: Study Buddy remembers where understanding breaks down and helps learners check that it sticks. This differentiation is a hypothesis to validate, not a claim that general-purpose assistants cannot perform these tasks.

2b. Top 3 MVP Value Props

The Vitamin: A clear explanation, source reference, and ten multiple-choice questions for a supported lesson.

The Painkiller: Immediate answer feedback identifies specific concepts to revisit.

The Steroid: Targeted reteaching and a focused retry respond to the student's actual mistakes, with weak concepts saved for later practice.

2c. Goals & Non-Goals

Goals

- Help beginner students complete a source-grounded lesson and check their understanding independently.
- Help students identify and revisit weak concepts through specific feedback and one targeted retry.
- Deliver trustworthy explanations and quizzes that stay within approved source material and the concepts actually taught.

Non-Goals

- Live web search, pasted notes, and file uploads are outside the first version; learning content comes only from an approved lesson library.
- Free-text grading, teacher dashboards, and unlimited quiz loops are deferred to keep the first build manageable.

2d. Success Metrics

Goal	Signal	Metric	Target
Complete a useful study session	Students finish a first quiz and view feedback	Completed first quizzes / started sessions in a 10-session pilot	At least 8 of 10 sessions; proposed target
Help students revisit weak concepts	Students improve after targeted reteaching	Compare first-attempt and retry scores, both out of 10, across at least 5 low-score study sessions	At least 60% of evaluated low-score sessions show an improvement of at least 1 point on the 10-question retry quiz; proposed target

Keep teaching and assessment grounded	Explanations and questions stay within approved material	Builder review of 50 questions across 5 lessons and their explanations	50/50 questions within taught scope; 0 unsupported teaching claims; pre-release target

2e. Expansion measurement plan

P1 evaluation: Track confidence-rating completion, correct/unsure and wrong/confident answers, and whether a suspected misconception is supported by a different question on the same concept. Review diagnostic labels against answer evidence; a single distractor is not proof of a misconception.

Track due-review completion and performance on new questions after a gap, including a proposed seven-day follow-up where feasible. Report returning-learner counts, timing, and baseline alongside results; an immediate retry score is not evidence of long-term retention. Set numerical P1 targets after a baseline pilot. The existing 60% retry-improvement target remains unchanged.

3. REQUIREMENTS

User Journey 1: A beginner student learns and practices a supported topic

Context: Turn a selected lesson into an understandable explanation and a short assessment, with feedback tied to what was taught.

Sub-journey: Choose a lesson and understand the topic

- **[P0] User can browse and select a topic from the approved lesson library.**
- **[P0] User can read a short beginner-friendly explanation generated by the Tutor (Agent 1) using only the selected approved lesson.**
- **[P0] User can see the lesson title, original source link, and key concepts alongside the explanation.**
- **[P1] User can choose a difficulty level supported by the selected lesson.**
- **[P2] User can search the approved library by topic name.**

Sub-journey: Check understanding with a quiz

- **[P0] User can answer exactly ten multiple-choice questions covering only concepts in the explanation.**
- **[P0] User can select one answer per question and submit the quiz before seeing the answer key.**
- **[P1] User can review and change selections before submitting.**
- **[P2] User can see how many questions remain unanswered.**

Sub-journey: Review feedback and practice missed concepts

- **[P0] User can view a score out of ten and feedback explaining each answer after submission.**
- **[P0] User can receive targeted reteaching and one focused 10-question retry quiz after a low score (below 8/10); after that retry, the session ends with a summary.**
- **[P1] User can see which concepts need more practice and follow the original lesson link to study further.**

User Journey 2: A returning student reviews progress and continues learning

Context: Preserve learning history so students can revisit previous work and focus later sessions on their weak concepts.

Sub-journey: Review previous study sessions

- **[P0] User can return to their saved quizzes, answers, feedback, scores, and weak concepts through a learner-scoped profile.**
- **[P0] User can see an accurate session status and a clear explanation if a technical failure prevents completion.**
- **[P1] User can view first-attempt and retry results separately for a lesson.**
- **[P2] User can view a simple score trend across completed sessions.**

Sub-journey: Continue with relevant practice

- **[P0] User can start another session informed by their saved weak concepts for a supported lesson.**
- **[P0] User can receive a clarification request when their input is necessary to resolve an ambiguous topic or choice.**
- **[P1] User can see a suggested next approved lesson or concept at the end of a session.**

User Journey 3: Understand mistakes and return for focused practice

Delivery: P1 expansion after the P0 session flow. Implement confidence ratings and misconception-tagged options first, followed by persistent concept history, scheduled review, and source-linked progress. No additional agent is required. These are planned requirements, not claims of implemented behavior.

Sub-journey: Capture confidence and understand a mistake

- [P1] For each quiz answer, the learner can choose Sure, Unsure, or Skip before submission and change that choice until submitting. Confidence is stored separately from correctness and never changes the deterministic score, pass threshold, or retry eligibility.
- [P1] Assessment (Agent 2) designs wrong options around plausible, lesson-grounded misconceptions. Each tagged option includes a concept ID, misconception ID, short description, and supporting source reference for controller validation. Tags are private until submission; they must not reveal the correct option in the learner quiz.
- [P1] After submission, the learner sees feedback adapted to correctness and confidence: correct/sure receives later review; correct/unsure receives reinforcement; wrong/unsure receives a simpler explanation; wrong/sure receives feedback addressing the possible misconception. Skipped confidence uses ordinary answer feedback.
- [P1] A wrong answer suggests a possible misconception, not a confirmed diagnosis. Confirm or challenge it with a different question on the same concept within an eligible 10-question retry or a later session. Never add an extra quiz loop merely to confirm a label.

Sub-journey: Continue practice across sessions

- [P1] A returning learner sees a concept-level history: what was practiced, mistakes and confidence, possible misconceptions, most recent evidence, and what to review next. Confidence is self-report and neither one correct answer nor a single high score establishes mastery.
- [P1] The controller maintains a due-review queue per learner and concept. Start with configurable intervals such as 1, 3, and 7 days; these are pilot defaults, not an optimal schedule claim. New answers update the next review date and the evidence shown to the learner.
- [P1] The learner can choose a due concept and start a new study session using approved material and saved evidence. Use new question wording or applications within taught scope; compare against stored question history to reject exact duplicates and flag near-duplicates for review.

- [P1] Every new practice session retains exactly 10 questions and the current limit of one focused 10-question learning retry after a first score below 8/10. Scheduling a future session never reopens a completed session or resets its retry budget. Review dates appear in the app; notifications are not required for P1.

Sub-journey: Understand progress and its sources

- [P1] Progress shows concept-level evidence alongside scores: for example, 'improved on membrane structure; cholesterol needs more practice.' Distinguish correct answers immediately after reteaching from correct answers in a later review.
- [P1] Feedback and progress decisions link to the approved lesson version and relevant passage or section, with attribution. If no stable section link exists, show the stored excerpt and original lesson link. A source link supports checking; it is not a guarantee of correctness.
- [P1] The learner can see why a concept was recommended, including the relevant attempt date and evidence. Use current-learner records only. Do not infer ability, intelligence, or clinical conditions from quiz answers.

4. APPENDIX

Project Repository

https://github.com/Soma-Majumder/Study-Buddy

Educational Content

Current approved lesson: Membrane Structure & Composition, version 2, adapted from Cell & Molecular Biology by Marg Sonnenfeld and Jonathan (Jung Lynn) Jung, Thompson Rivers University (https://cellandmolecularbiology.pressbooks.tru.ca/chapter/2-1/), under CC BY-NC-SA 4.0 (https://creativecommons.org/licenses/by-nc-sa/4.0/). Preserve attribution, the adaptation notice and the same-license/noncommercial conditions. OpenStax remains a possible future source subject to per-lesson review.

Before importing any lesson:

- Review each selected lesson’s license
- Keep source links and attribution
- Approve the content

Biology is the current starting subject; one lesson is approved. Four additional curated lessons are still needed for the proposed five-lesson pilot.

Multi-Agent Workflow

Student selects a topic

↓ Controller starts the session

↓ Tutor (Agent 1) retrieves and simplifies the approved lesson

↓ Controller validates the explanation, source references, and concept IDs

↓ Assessment (Agent 2) builds a private 10-question candidate and key

↓ Controller validates the contract; Reviewer checks grounding and question quality

↓ If required, Assessment revises once and Reviewer checks the replacement; unresolved blocking issues fail explicitly

↓ Controller publishes only the exact approved candidate; nonblocking quality warnings remain visible in backend metadata

↓ Controller shows the quiz and waits for student answers

↓ Student submits answers

↓ Controller grades submitted answers against the saved key

↓ Controller derives weak concepts from verified answers; Assessment (Agent 2) explains those results

↓ Controller checks: score below 8/10 AND learning retry unused?

Yes → Controller validates weak-concept handoff → Tutor (Agent 1) reteaches → controller validates revised explanation → Assessment (Agent 2) builds one focused 10-question retry through the same review/revision/publication gate → student submits → controller grades → Assessment (Agent 2) explains results

No → continue

↓ Controller saves progress, shows summary, and only then marks the session complete

Shared state: learner/session ID, lesson ID/version, explanation ID, quiz/attempt IDs, concept IDs, answers, verified score, feedback, weak concepts, learning retry count, and repair counts keyed by handoff ID. Only controller code writes validated records.

Every agent handoff includes task, relevant context, sources, expected output, and uncertainty with reasons.

Critical handoff: Tutor (Agent 1) → controller → Assessment (Agent 2). Missing explanation or source/concept references blocks quiz creation; allow one repair, then fail explicitly.

Return handoff: Assessment (Agent 2) → controller → Tutor (Agent 1). Missing weak concepts blocks reteaching; allow one repair for that handoff, then fail explicitly. Validate concept IDs against the current lesson and quiz attempt before reteaching.

Agreed Rules

Roles

- **Tutor (Agent 1) teaches**
- **Assessment (Agent 2) creates/revises quizzes and explains controller-verified results**

- **Reviewer checks candidate quality and returns evidence-bound findings; controller code decides approval and publication**

Handoffs

- Every handoff includes: **task, context, sources, expected output,** and **uncertainty with reasons**
- Application code controls the sequence and validates handoffs
- An incomplete handoff gets at most one technical repair, then explicit failure. Tutor and Assessment share the outgoing handoff repair budget. Content revision has a separate one-revision budget; malformed Reviewer reports fail without a Reviewer repair loop.

Students & Quizzes

- Ask students only when their input is required
- Quizzes test **taught concepts only** and use **multiple choice**
- Low scores trigger **one focused retry** after reteaching

What gets saved: quizzes, answers, feedback, scores, and weak concepts

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Proposed Implementation Details

- **Pass threshold: 8/10**
- **Grading:** The controller checks answers against the saved key *deterministically*
- **Explaining: Assessment (Agent 2) explains controller-verified results and weak concepts; controller code owns correctness, score and missed-concept counts**
- **Answer keys:** Kept on the backend until submission
- **Retry budgets: Technical repairs, content revision and learning retries have separate persisted budgets. Each quiz-quality operation allows at most five provider reservations, one content revision and two reviews, within a 180-second deadline; each call is capped at 60 seconds and the remaining allowance. Malformed Reviewer reports fail explicitly. Quality policy: six distinct understanding questions is the generation goal; shortfall and repeated-target findings are warnings, with conditional quiz-level cue warnings when no individual question fails. Blocking question defects prevent publication. This does not change the 8/10 pass threshold or create another student retry.**
- Technical failures **never** count as wrong answers

P1 role ownership and handoff extensions

Controller: Grades deterministically, derives weak concepts from verified answers, combines correctness with self-reported confidence, validates option tags and source references, records diagnostic evidence, schedules reviews, and chooses the next eligible operation. It assembles the return handoff and atomically reserves a learning retry for the current session/attempt before calling Tutor. Duplicate requests return the existing operation status; they do not start another retry.

Assessment (Agent 2): Creates in-scope questions and provisional misconception-tagged distractors, explains controller-verified results, and proposes source-linked feedback. It does not decide the score, assert mastery, confirm a misconception from one answer, or write learner state.

Tutor (Agent 1): Receives only the validated focus concepts, relevant misconception evidence and confidence context, and approved source excerpts. Reteaches those ideas in simple language. The private answer key is not included.

Controller to Assessment: Include lesson/version, current handoff and explanation IDs, taught concept IDs, the explicit question count, and relevant prior question references to avoid repeats. Assessment returns per-option diagnostic tags and source references as private assessment metadata.

Assessment through controller to Tutor: Preserve task, session/handoff IDs, lesson/version, quiz/attempt IDs, verified score and missed counts, sources, expected output, and uncertainty. P1 adds selected-option evidence, optional confidence, provisional misconception IDs/status, and whether supporting evidence is present. Correct-answer feedback may reinforce uncertainty without opening an additional learning retry.

Version and validation: Define a versioned protocol extension before implementing P1; do not send new fields to the current strict v1 schemas. Reject stale sessions, unknown concepts/tags, invalid confidence values, and mismatched sources. If confidence was skipped or tags are unavailable, keep the valid score and provide ordinary feedback without inventing a diagnosis. Missing required weak concepts still follows the one-repair-then-explicit-failure rule.

Data Model

Table	What it stores
lessons	topic, source text, URL, attribution, version
study_sessions	learner, lesson, status, learning retry count, repair counts by handoff ID
explanations	text, concept IDs
quizzes & questions	rounds, private keys
attempts & responses	answers, scores, feedback
weak concepts	learner ID, concept IDs, source attempt ID, lesson ID, recorded_at

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P1 data-model extensions (planned)

Responses: learner-scoped attempt/question IDs, selected option, is_correct, confidence (sure, unsure, or null), and submitted_at. Learner confidence is separate from agent uncertainty in the handoff.

Private option diagnostics: question ID/version, option ID, concept ID, optional misconception ID and description, source lesson/version and passage reference, and validation status. Correct options have no misconception tag.

Concept evidence: learner ID, concept ID, source attempt/question, correctness, confidence, provisional misconception ID/status, evidence timestamp, and source references. Preserve attempts separately; do not turn a transient score into a permanent mastery label.

Review schedule: learner/concept ID, lesson/version, last-practiced time, next-review time, interval stage, supporting attempt IDs, and schedule-policy version. Use a unique active learner/concept schedule and idempotent updates to avoid duplicate due tasks.

Question history and sources: learner/session/question IDs, question text or fingerprint, tested concept, round, used_at, approved passage ID/text, source URL/section, attribution, and lesson version. Preserve historical evidence on its original version; revalidate material before scheduling a revised lesson.

Tool Boundaries

Who	Can do
Tutor (Agent 1)	`find_lesson`, `get_lesson`
Both agents	`get_previous_attempts`, scoped to the current learner only
Controller code only	Write validated session and progress records

No arbitrary access to another learner's history.

Failure Handling

- Treat embedded instructions in source text as **data, not commands**
- **Reject** unsupported teaching or questions outside the explanation
- If any of these happens:
  - an unresolved handoff
  - a missing source
  - a failed save
- → Show a **useful error** and record the **actual status**. Never claim success.

Acceptance Checks *(run before release)*

Test 1: Perfect score

- Supported lesson, all answers correct
- Ten in-scope questions → 10/10
- Feedback saved, session completes **without retry**

Test 2: First score 6/10

- Reteach the missed concepts
- Generate one focused ten-question retry
- Wait for submission and save **both** attempts
- End regardless of the retry score

Test 3: Empty source or malformed handoff

- **One repair** at most

- No invented content or score
- Explicit failure if unresolved

Test 4: Borderline pass — first attempt 8/10

- **Passes; no reteaching or retry. Save results and show summary.**

Test 5: Just below threshold — first attempt 7/10

- **Triggers targeted reteaching and one focused 10-question retry when unused. After the retry, save progress and show a summary before marking the session complete regardless of score.**

Test 6: Missing weak concepts

- **Assessment (Agent 2) returns no weak concepts after a low score.**
- **Allow one repair at most; if weak concepts remain missing, fail explicitly and do not start reteaching.**

P1 acceptance checks

- Confidence matrix: Test all four correctness/confidence combinations and skipped confidence. Scores and the 8/10 threshold remain identical regardless of confidence; no extra learning retry is created.
- Misconception evidence: A wrong/confident answer gets a provisional label only. A different question can support or contradict it; untaught concept tags, unsupported source references, and prematurely exposed private metadata are rejected.
- Return visit: Saved evidence survives a new session for the same learner. A due concept produces new in-scope practice; the original session stays completed with its original retry count. Another learner cannot read the history.
- Schedule/concurrency: Simulated dates produce the configured due reviews without real waiting. Replayed updates and simultaneous retry requests create one schedule update or one reserved retry, respectively.
- Source-linked progress: Each displayed recommendation resolves to the correct learner, attempt, concept, lesson version, and stored passage. Stale or missing references produce an explicit unavailable state rather than fabricated evidence.
- Scope and retention: Every initial/retry quiz still has 10 questions. Test duplicate-question rejection, unchanged save-failure behavior, and separate reporting of immediate retry versus later review performance.

Also verify:

- Learner isolation
- Answer keys are absent from pre-submission client payloads
- A failed save shows an error and records the actual failed or pending-save status rather than complete.

Open Decisions

- Additional pilot lessons beyond the approved cell-membrane lesson
- Implemented provider: Claude through the Anthropic API; model configuration remains configurable and live content quality still requires evaluation

- Implemented storage: SQLite. Production hosting, process supervision, backups and deployment ownership remain to be decided
- Learner identity method
- History retention
- Delivery schedule

Note: Metrics are proposed targets, not observed results. Offline workflow tests and limited development evaluations exist; a student pilot and independent live Reviewer validation have not been completed.